

COS10005

Web Development

Lecture 4

HTML Structure, Content Models & CSS

Contents

- HTML Structure and content model
- HTML structure elements: Main, Article, Section, Header
- HTML layout structure: Division, Navigation, Span
- History of HTML and Cascade Style Sheets (CSS)
- Benefits of using CSS
- Selector and property
- Different ways to format a page

Page Structure

- Division and Span
- HTML5: Article, Section, Header, Main, Footer, and Navigation
- HTML5: Aside and Figure

HTML Common Structure Elements

1. **Headings:** Headings help to organize the content into sections, with the main heading (H1) usually being the most important and subsequent headings (H2, H3, etc.) providing a hierarchy of sub-sections.
2. **Navigation:** Navigation provides a way for users to move around the website, with links to other pages, sections or categories.
3. **Content:** The main content of the page should be organized in a logical way, with related content grouped together and separated by whitespace.
4. **Footer:** The footer often contains information such as copyright contact information and links to related content.

HTML Content Model

- Different types of content may require different HTML tags, and understanding content models can help you to create more semantic and accessible web pages. Here are some common content models:
 1. Text: Text content includes headings, paragraphs, and other text-based elements such as lists, blockquotes, and tables.
 2. Images: Images can be included on web page using the `<img>` tag, and can be given a descriptive alt attribute to improve accessibility.
 3. Multimedia: Multimedia content such as videos and audio can be included using the `<video>` and `<audio>` tags.
 4. Forms: Forms allow users to submit information to a website, and can include input elements such as text boxes, radio buttons, checkboxes and dropdown lists.

HTML Structure - Main

- `<main> .. </main>` represents the main content of the `<body>` of a web page.
- It is used to wrap the content that is central to the page purpose, and is separate from any content that is repeated across multiple pages such as headers, footer, or navigation.
- Only one `<main>` is permitted in a `<body>`.
- The main content area consists of content that is directly related to or expands upon the central topic of a document or central functionality of an application.

HTML Structure - Article

- `<article>` .. `</article>` defines independent, self-contained content that can be independently distributed or reused.
- It is typically used to wrap content that could stand alone and the meaningful on its own, such as blog posts, news articles, or product reviews.
- Content should make sense on its own and it should be possible to distribute it independently from the rest of the site.
- Example are:
 - Forum post
 - Blog post
 - News story
 - Comment

Example of Article

```
<header>
  <nav>
    <!-- navigation links go here -->
  </nav>
</header>
<main>
  <article>
    <h1>10 Tips for Better Sleep</h1>
    <p>If you're having trouble getting a good night's sleep, try
these tips:</p>
    <ul>
      <li>Stick to a consistent sleep schedule</li>
      <li>Create a relaxing bedtime routine</li>
      <li>Avoid caffeine and alcohol before bed</li>
      <li>Make your bedroom dark and quiet</li>
      <li>Exercise regularly, but not before bedtime</li>
      <li>Don't eat heavy meals before bedtime</li>
      <li>Avoid naps during the day</li>
      <li>Limit your screen time before bed</li>
      <li>Try relaxation techniques like deep breathing or
meditation</li>
      <li>Consider talking to a doctor or sleep specialist</li>
    </ul>
  </article>
</main>
```

10 Tips for Better Sleep

If you're having trouble getting a good night's sleep, try these tips:

- Stick to a consistent sleep schedule
- Create a relaxing bedtime routine
- Avoid caffeine and alcohol before bed
- Make your bedroom dark and quiet
- Exercise regularly, but not before bedtime
- Don't eat heavy meals before bedtime
- Avoid naps during the day
- Limit your screen time before bed
- Try relaxation techniques like deep breathing or meditation
- Consider talking to a doctor or sleep specialist

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HTML Structure - Section

- `<section>` .. `</section>` defines sections in a document. Such as chapters, headers, footers, or any other document sections.

```
<body>
  <header>
    <nav>
      <!-- navigation links go here -->
    </nav>
  </header>
  <main>
    <section>
      <h2>About Us</h2>
      <p>We are a family-owned business that has been serving the c
    </section>
    <section>
      <h2>Our Products</h2>
      <ul>
        <li>Widgets</li>
```

HTML Structure - Header

- `<header> .. </header>` defines a header for a document, article and section.
- It should be used as a container for introductory content or a set of navigational links for its content.
- There can be several `<header>` elements in one HTML document.
- Note: It cannot be placed within a `<footer>` or another `<header>` element.

HTML Structure - Footer

- `<footer> .. </footer>` defines a footer for a document or a section.
- It should contain information about its **containing** content, such as copyright information, links to terms of use, contact information.
- Location information can be enclosed in `<address> .. </address>` tags.
- There can be several `<footer>` elements on one web page.

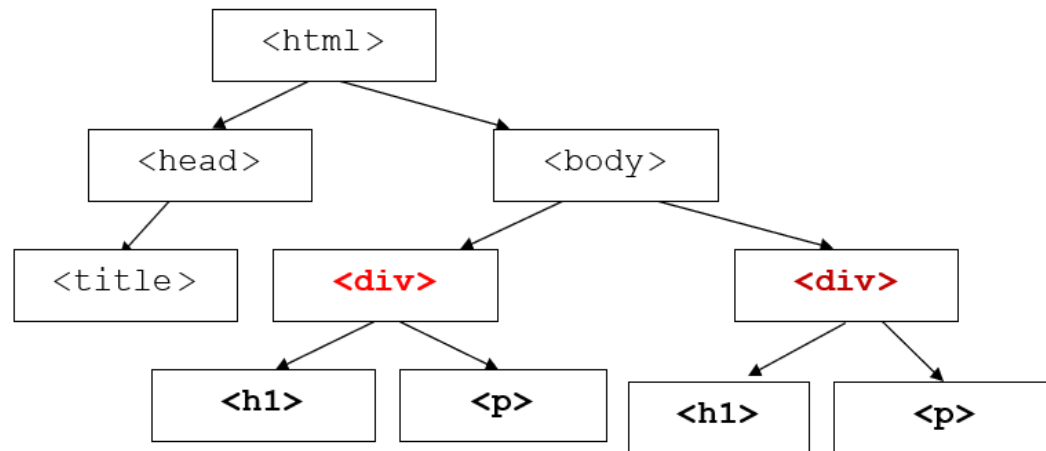
Footer Example

```
<body>
  <article>
    <header>...</header>
    <section>...</section>
    <footer>
      Posted by: John Smith 27 Aug
    </footer>
  </article>
  <article>
    <header>...</header>
    <section>...</section>
    <footer>
      Posted by: Derp Derpina 19 March
    </footer>
  </article>
</body>
```

Division

- The **html** element – the ‘**root**’ element of any html document usually contains only two children **head** and **body**.
- The **head** then contains the **title**, and other ‘head’ elements.
- The **body** can contain many other elements, grouped by **div** elements.

```
<html>  
<head>  
  <title>...</title>  
</head>  
<body>  
  <div>  
    <h1>...</h1>  
    <p>...</p>  
  </div>  
  <div>
```



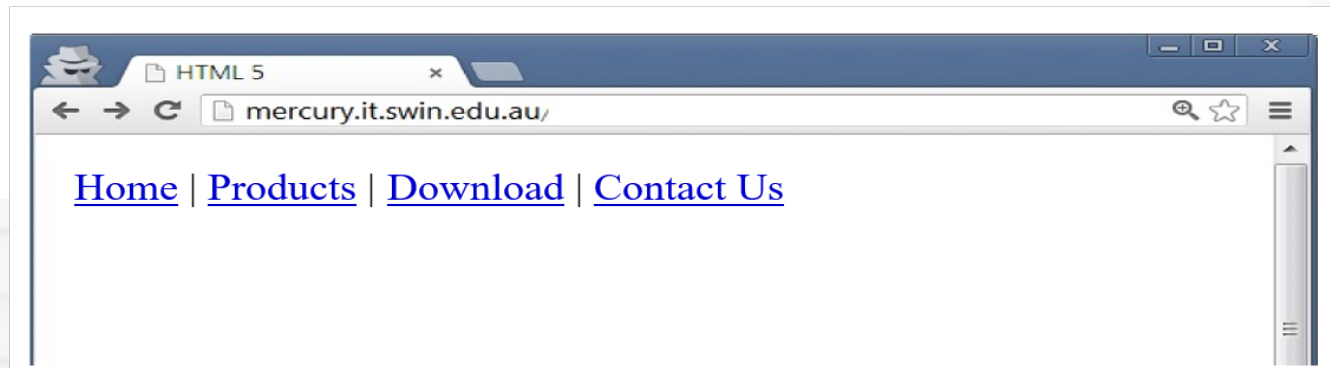
Navigation

- `<nav> .. </nav>` specifies a section of navigation links.
- It is intended only for major block of navigation links.
- There can be **multiple** `<nav>` elements on one web page.

`<nav>`

```
<a href="index.php">Home</a> |  
<a href="product.php">Products</a> |  
    <a href="download.php">Download</a> |  
    <a href="contact.php">Contact Us</a>
```

`</nav>`



HTML Structure: `<span>`

- `<span>` .. `</span>` is a generic **inline** level container used to group other inline elements, such as text.
- Similar to **the** `<div>` elements, the `<span>` tag is used to group content, but only for inline content.
- **Do not** use a `<span>` when you should be using a logical element like `<em>` or `<strong>`.

HTML5 - Figure

- `<figure> .. </figure>` encloses a self-contained content such as illustrations, diagrams, photos, code listings.
- Its position should be independent of the main flow, and can change without affecting the flow of the document.
- The `<figcaption>` element is used to add a caption for the `<figure>` element.

HTML5 - Figure

```
<p>The Advanced Technologies  
Centre (ATC) has achieved a 5  
Star Green Star - Education  
Design v1 rating from the  
Green Building Council of  
Australia.</p>
```

```
<figure>
```

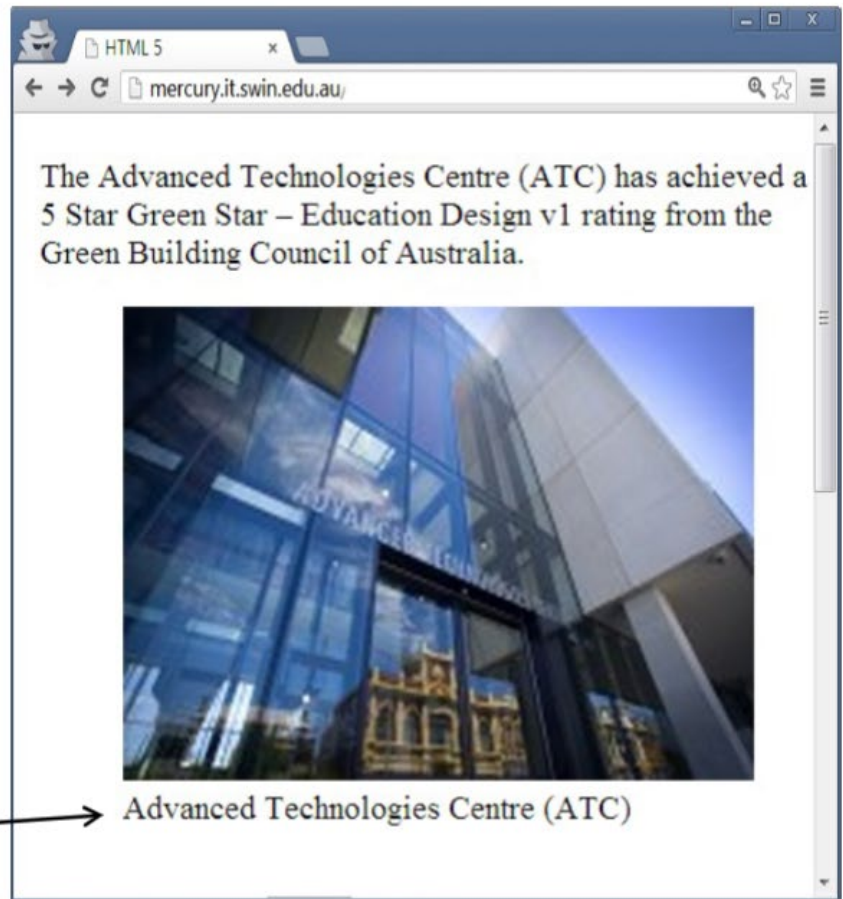
```
  
```

```
  <figcaption>
```

```
    Advanced Technologies Centre  
    (ATC)
```

```
  </figcaption>
```

```
</figure>
```



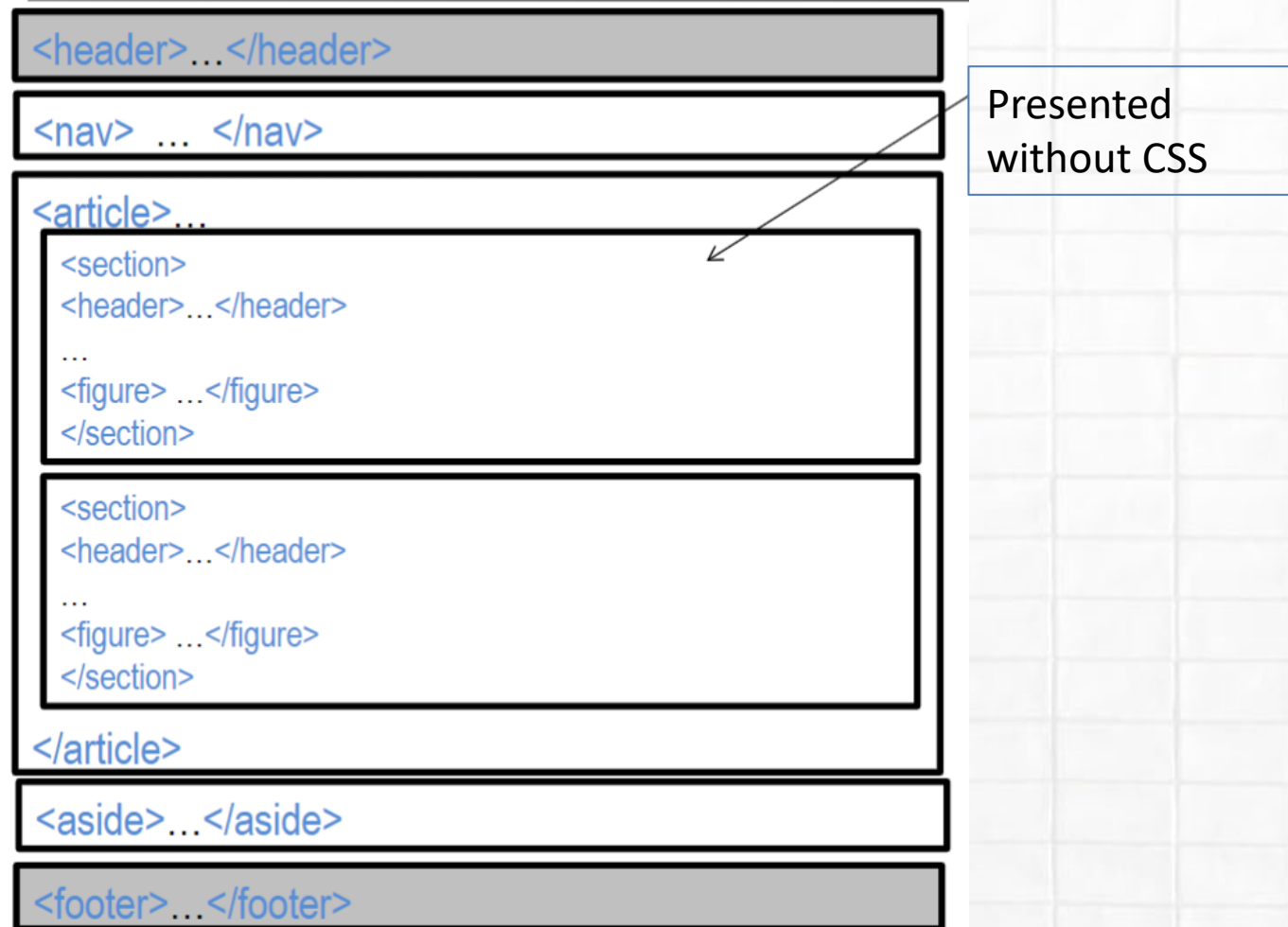
HTML5 - Aside

- The `<aside> .. </aside>` element is used to contain supporting information related to the surrounding content, such as related reading links and glossaries.

```
<p>This semester I'm taking up COS10005.</p>
<aside>
<h4>COS10005</h4>
<p>COS10005 is a web development subject
covering HTML, CSS and JavaScript.</p>
</aside>
```



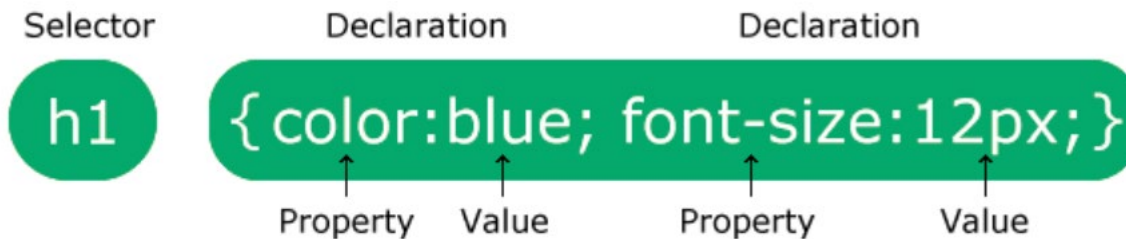
Example of Web Page Structure



What is CSS

- CSS stands for Cascading Style Sheets.
- CSS is a styling language used to describe the visual presentation of a web page written in HTML.
- It's used to change the font, color, spacing, layout, and other visual aspects of the content.
- It can control the layout of multiple web pages all at once.
- External stylesheets are stored in CSS files.

CSS Format



- H1 is a selector in CSS (it points to the HTML element you want to style).
- Color is a property, and blue is the property value.
- Font-size is a property and 12px is the property value.

CSS Selectors

- **CSS Selectors:** are used to target HTML elements and apply styles to them.
- There are different types of selectors, such as element selectors, class selectors, and ID selectors.
 - **Element selector:** targets all instances of a particular HTML element. For example, to style all paragraphs in a document, you could use the following selector:

```
p { color: red; }
```

CSS Selectors

- **CSS Selectors:** targets all instances of an element with a particular class. For example, to style all elements with the class “highlighted”, you could use the following selector are used to target HTML elements and apply styles to them.

```
.highlighted {  
  background-color: yellow;  
}
```

CSS Selectors

- **ID Selectors:** targets a single element with a particular ID. IDs should be unique within a document. For example, to style the element with the ID “header”, you could use the following selector:

```
#header {  
  font-size: 24px;  
}
```

Selector Notations

- The “#” symbol is used to select elements based on their ID.
- The “.” symbol is used to select elements based on their class.

```
<div id="header">This is the header</div>
```

CSS

```
#header {  
  background-color: blue;  
  color: white;  
}
```

```
<div class="box">This is a box</div>  
<div class="box">This is another box</div>
```

CSS

```
.box {  
  border: 1px solid black;  
  padding: 10px;  
  margin: 5px;  
}
```

Selector Notations

- The “>” symbol is used to select child elements.
- The first CSS rule applies only to the <h2> element that is a direct child of the element with the ID “container”, and the second CSS rule applies only to the elements that are direct children of the element that is a direct child of the element with the ID “container”.

html

```
<div id="container">
  <h2>Title</h2>
  <ul>
    <li>Item 1</li>
    <li>Item 2</li>
  </ul>
</div>
```

CSS

```
#container > h2 {
  font-size: 24px;
}
#container > ul > li {
  color: blue;
}
```

CSS Properties

CSS properties define the visual styles to be applied to HTML elements. There are many properties available, such as font-size, color, background-color, and margin. Here's an example of how to use some properties:

```
p {  
  font-size: 16px;  
  color: #333;  
  background-color: #f8f8f8;  
  margin-bottom: 20px;  
}
```

CSS Properties

The CSS box model describes how elements are sized and positioned on a web page.

Every HTML element is represented as a rectangular box, with four properties: margin, border, padding, and content. Here's an example:

```
.box {  
  width: 200px;  
  height: 200px;  
  padding: 20px;  
  border: 1px solid #333;  
  margin: 10px;  
}
```

CSS Layout

CSS can be used to create different layouts for web pages. There are several techniques for achieving layout, such as using floats, flexbox, and grid. There's an example of how to use flexbox:

```
.container {  
  display: flex;  
  flex-wrap: wrap;  
  justify-content: center;  
}  
  
.item {  
  flex-basis: 200px;  
  margin: 10px;  
}
```

CSS Example

- Change the font size and family of a paragraph element:

```
p {  
  font-size: 16px;  
  font-family: Arial, sans-serif;  
}
```

- Add a border and padding to an image element:

```
img {  
  border: 1px solid black;  
  padding: 10px;  
}
```

Wake brain

- Change the color and background of a heading element?



```
h1 {color: red;  
    background-color: yellow;}
```



```
<h1 style="background-color:yellow"> This is a  
header </h1>
```



All Above



None Above

Wake brain

- Which of the following CSS selectors targets all the <p> elements that are descendants of the element with the ID “content”?



#content p



p#content



content>p



p.content

CSS Animation and Transition

- CSS can be used to create animations and transitions on web pages. This adds interactivity and visual interest.
- Animations involve changing the CSS properties of an element over time.
- Transitions involved smoothly transitioning between two CSS states. Here's an example

```
/* Animate a button when it's hovered over */  
button {  
    transition: all 0.2s ease-in-out;  
}  
  
button:hover {  
    transform: scale(1.1);  
    background-color: #333;  
    color: white;  
}
```

Wake Brain

What is the purpose of using semantic HTML structural elements, such as `<header>`, `<footer>`, `<nav>`, `<article>`, and how can they improve the accessibility, SEO, and user experience of a web page?



Screen readers can use the `<header>` and `<footer>` elements to navigate a web page more easily, and users with disabilities can better understand the layout and hierarchy of the content.



SEO: By providing more context and meaning to search engines these elements can improve SEO. For example, search engines can use the `<article>` element to identify and prioritize the main content of a page, which can improve its ranking in search results.



User experience: They improve the overall user experience of a web page by making it easier to navigate and understand. For example, the `<nav>` element can be used to create a consistent and easy-to-use navigation menu.



Thank You